

Week 1: What is (Spatial) Statistics?

1.1. Mathematical Underpinnings

Vector	-array of numbers -row or column vectors -each element can represent either each dimension or entity -n-dimensional vector directs to a certain point in n-dimensional space
Matrix	-two dimensional representation of numbers -each dimension is assigned to each column -each data point, instance, case, or individual is aligned along rows -types of matrix • square • diagonal: entries outside the main diagonal are all 0 • identity (I): 1s in the diagonal, 0 otherwise

-elementary matrix calculation

- let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$

Addition	Subtraction	Multiplication	Hadamard Product
$A+B = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$	$A-B = \begin{bmatrix} a-e & b-f \\ c-g & d-h \end{bmatrix}$	$AB = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$	$A \odot B = \begin{bmatrix} ae & bf \\ cg & dh \end{bmatrix}$

1.2. Level of Measurement

-types of measurement scales

	Definition	☞	Summaries
Nominal	-categories -without any intrinsic ordering	-gender, type of a car, dog names	-frequency tables -mode -bar/pie charts
Ordinal	-categories -with a ranking or order -no consistent difference between ranks	-satisfaction levels (Likert Scale)	-frequency distributions -median (if categories are numeric & can be ranked) -bar charts
Interval	-numeric scale -meaningful differences -no true zero point	-temperature in Celsius	-measures of central tendency: mean, median, mode -measures of dispersion: variance, standard deviation
Ratio	-numeric scale -meaningful distances -true zero	-weight, height, distance, temperature in Kelvin unit	

1.3. Descriptive Statistics

-two versions

- **1**: $n \times 1$ vector of all 1s

	Arithmetic	Vector
Sum	$\text{sum}(X) = \sum_{i=1}^{ X } x_i$	
Mean	$\text{mean}(X) = \frac{1}{ X } \sum_{i=1}^{ X } x_i$	$\bar{x} = \frac{1}{n} \mathbf{1}^\top \mathbf{x}$
Variance	$\text{variance}(X) = \frac{1}{ X } \sum_{i=1}^{ X } (x_i - \text{mean}(X))^2$	$\sigma_x^2 = \text{Cov}(\mathbf{X}, \mathbf{X}) = \frac{1}{n} (\mathbf{x} - \bar{x}\mathbf{1})^\top (\mathbf{x} - \bar{x}\mathbf{1})$
Covariance		$\text{Cov}(\mathbf{X}, \mathbf{Y}) = \frac{1}{n} (\mathbf{x} - \bar{x}\mathbf{1})^\top (\mathbf{y} - \bar{y}\mathbf{1})$

- median: the value at the center when all values are ordered by magnitude
- mode: the most frequent value in a given set

Week 2: Correlation and Summary Statistics

2.1. Revisiting Matrices

-2 dimensions in a matrix $\rightarrow$ matrix can be seen from each dimension

Row-centered look	Column-centered look
-each row represents an individual (instances/point/area) -each individual may have multiple attributes(변수) -it is manipulated to obtain a mathematically sound form for analysis $\Rightarrow$ computing inverse	-each column represents a variable -don't confused with the term column vector of which elements represent dimensions (matrix consists of stacked transposed column vectors) -each column can be assumed to be generated by a process -cross-column relationship is important in statistical modelling

-vector and matrix multiplication

Dot-product	-a sum of the products of corresponding elements in two vectors -the result is scalar (dimensionless) -self dot-product gives the sum of squares $\mathbf{x} \cdot \mathbf{x} = \mathbf{x}^T \mathbf{x}$ $= \sum_{i=1}^{ \mathbf{x} } (x_i \times x_i)$
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2.2. Correlation and Covariance

Correlation	-measures linearity in the relationship between two variables -normalized measure of dependence -one of the most commonly used correlation coefficients: Pearson's r $\rho_{XY} = \frac{Cov(X, Y)}{\rho_X \rho_Y}$ • ranges from -1 (perfect negative correlation) to 1 (perfect positive correlation) • if $\rho = 0$: no linear association between the two variables -rank correlation <table border="1"> <tr> <td>Spearman's ρ</td><td> -replace X with $R(X)$ in the formula of Pearson's r </td></tr> <tr> <td>Kendall's τ</td><td> -consider pairs of values (순서쌍) -concordant (부합) and discordant (비부합) pairs are distinguished • concordant: the other chosen data forms upward line when connected • discordant: the other chosen data forms downward line $\tau_{XY} = \frac{ [\text{concordant pairs}] - [\text{discordant pairs}] }{ [\text{all pairs}] }$ -note that $[\text{all pairs}] = \binom{n}{2} = \frac{n(n-1)}{2}$ </td></tr> </table>	Spearman's ρ	-replace X with $R(X)$ in the formula of Pearson's r	Kendall's τ	-consider pairs of values (순서쌍) -concordant (부합) and discordant (비부합) pairs are distinguished • concordant: the other chosen data forms upward line when connected • discordant: the other chosen data forms downward line $\tau_{XY} = \frac{ [\text{concordant pairs}] - [\text{discordant pairs}] }{ [\text{all pairs}] }$ -note that $[\text{all pairs}] = \binom{n}{2} = \frac{n(n-1)}{2}$
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Covariance	-measures how two variables X and Y change together $Cov(X, Y) = E[(X - E[X])(Y - E[Y])]$ -if X and Y are independent, then $Cov(X, Y) = 0$				

-measuring linearity

- correlation coefficient can only give information whether two variables are **linearly** correlated
- converted or transformed variables may result in high correlation when such variables are getting linearly dependent

-note on summary statistics

- summary statistics provide succinct and useful information
- however, identical summary statistics do not always guarantee that the underlying distribution or trends in values are the same

2.3. Quantiles

Quantiles	-values that splits the data into equal-sized portions -tertile / quartile / quintile / decile / percentile -mathematical formula for a quantile Q_p $Q_p = \min\{x_i F(x_i) \geq p\}$ • where $F(x)$ is the cumulative distribution function • means that “at least” 25% (Q1) / 75% (Q3) of the values are below Q_p
Quartiles	-50th percentile = 2nd quartile (Q2) = median -interquartile range (IQR) measures the spread of the middle 50% of data (IQR = Q3 – Q1) • convenient means of identifying variability • resistant to outliers -box-whisker plot • box: spanning Q1 to Q3 • whiskers: extending $1.5 \times \text{IQR}$ from Q1 and Q3 i.e. $Q1 - 1.5 \times \text{IQR}$ / $Q3 + 1.5 \times \text{IQR}$ • median line, outliers (points beyond whiskers)

2.4. Weighted Statistics

-in some cases, different observations carry different importance $\Rightarrow$ modeled using weights

-notations

	Arithmetic	Vector
Weighted Sum	$\text{sum}_w(X) = \sum_{i=1}^{ X } w_i x_i$	
Weighted Mean	$\text{mean}_w(X) = \bar{X}_w = \frac{1}{\sum_{i=1}^{ X } w_i} \sum_{i=1}^{ X } w_i x_i$	$\bar{x}_w = \frac{1}{\sum_{i=1}^n w_i} \mathbf{w}^\top \mathbf{x}$
Weighted Variance	$\text{variance}(X) = \frac{1}{\sum_{i=1}^{ X } w_i} \sum_{i=1}^{ X } w_i (x_i - \bar{X}_w)^2$	$\sigma_w^2 = \frac{1}{\sum_{i=1}^n w_i} \mathbf{w} \odot [(\mathbf{x} - \bar{x}_w \mathbf{1})^\top (\mathbf{x} - \bar{x}_w \mathbf{1})]$
Covariance		$\text{Cov}(\mathbf{X}, \mathbf{Y}) = \frac{1}{\sum_{i=1}^n w_i} \mathbf{w} \odot [(\mathbf{x} - \bar{x}_w \mathbf{1})^\top (\mathbf{y} - \bar{y}_w \mathbf{1})]$

-may replace the sum of weights to $\sum_{i=1}^n w_i = (\mathbf{w}^{1/2})^\top \mathbf{w}^{1/2}$ where $\mathbf{x}^{1/2} = [\sqrt{x_1}, \dots, \sqrt{x_n}]^\top$

Week 3: Probability Distributions and Hypothesis Testing

3.1. Foundations

-sample space(Ω) contains all possible outcomes

-event is any subset of the sample space

-probability of an event E : $P(E) = \frac{\text{number of outcomes in } E}{\text{total number of outcomes in the sample space}}$

Probability Axioms	-P satisfies axioms • $P(E) \geq 0$ • $P(\Omega) = 1$, where Ω is the sample space. • $P(\cup_i E_i) = \sum P(E_i)$ for disjoint events (배반 사건) E_i
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-permutation: ordering or arrangement of objects

- if you have n distinct objects, the total number of ways to order them is $n!$
- if you only arrange k out of n items (without replacement), the number of permutation is

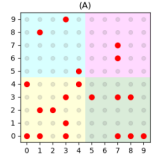
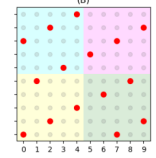
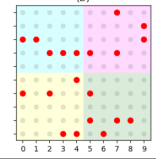
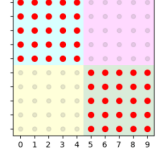
$$P(n, k) = \frac{n!}{(n-k)!} = k! \frac{n!}{(n-k)!} = k! \binom{n}{k}$$

3.2. Population and Samples

-objective: get sound estimates of summary, distributional properties of the population from sample(s)

-we always consider cases where samples are only available

-sampling methods: guiding the way we collect parts of the population

Simple Random Sampling	-equal chance for each member -unbiased, easy to interpret -might not capture subgroups adequately	
Systematic Sampling	-select every k-th item -straightforward to implement -can be biased if there is a hidden periodic or regular pattern	
Stratified Sampling	-partition population into strata and sample within each -high precision if strata are homogeneous -heavily relying on stratification	
Cluster Sampling	-make clusters then randomly select some of those -cost-effective (no references are required) -subject to sampling error if clusters differ	

-related techniques

Monte Carlo Methods	-simulate from a known distribution to approximate integrals or solve problems -flexible, can handle complex distributions -purely numerical, can be slow for high accuracy
Permutation (Bootstrapping)	-resample from existing data to assess variability -often with replacement -non-parametric, minimal assumptions -relies on the degree of sample representing the population

-sample statistics: summary statistics for samples

- mean: $\bar{x} = \frac{1}{n} \sum x_i$
- variance: $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$
- standard deviation: $s = \sqrt{s^2}$
- covariance: $q_{xy} = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$

-law of large numbers

- as you increase sample size, sample means converge to the population mean
- this extends to other sample statistics for i.i.d samples

3.3. Statistical Hypothesis Testing

Frequentist	-truth (population parameter) is fixed but unknown -probability is long-run frequency -hypothesis testing interprets data assuming a model; uses p-values and confidence intervals
Bayesian	-start from a prior distribution on the truth -probability is “degree of belief” -posterior distribution updates prior knowledge with data likelihood, giving a full distribution of parameters

-key differences

- frequentist: parameters are fixed, data are random
- bayesian: parameters are random, data are observed

-mechanics of frequentist hypothesis rejection

- setup: null hypothesis (H_0) vs. alternative hypothesis (H_1)
- one-sided vs two-sided
- rejection criterion: compute test statistic, find p-value $\Rightarrow$ if $p \leq \alpha$ (e.g. 0.05), reject H_0 or fail to reject

-p-hacking: inappropriate practices to show the significance

-p-value: the probability under the null hypothesis of observing a test statistic at least as extreme as the one actually observe

- rejecting H_0 does not prove H_1 : it only indicates the data are “unlikely” if H_0 were true
- non-significant result does not prove H_0 is true: it might be insufficient evidence or small sample size
- p-values can be misinterpreted as “probability that H_0 is true” $\Rightarrow$ they reflect how extreme the data are under H_0
- even under true H_0 , p-values fall below 0.05 about 5% of the time

-error types

	H_0 True	H_0 False
Reject H_0	Type I error	Correct Rejection
Fail to reject H_0	Correct Decision	Type II error

Type I Error	-rejecting the null hypothesis (H_0) when it's actually true (false positive, 거짓 양성)
Type II Error	-failing to reject the null hypothesis when it's actually false (false negative, 거짓 음성)

-experiment

- effect: mean shift in group 2 from group 1 with zero mean $\Rightarrow$ two groups are simulated following $N(0,1^2)$ and $N(\text{effect},1^2)$
- 30 samples are drawn from two groups 10,000 times and measure type I and type II errors
- case A: effect = 0 (H_0 is true) $\Rightarrow$ 520 Type I errors
- case B: effect = 0.5 (H_0 is false) $\Rightarrow$ 5219 Type II errors

3.4. Major Statistical Distributions

-data-generating process could be represented as a model form

-models are often controlled by parameters: parameters control the shape and location of the distribution and constitute a function (pmf, pdf)

-popular distributions

Normal/Gaussian	μ (mean) and σ (standard deviation)	$N(\mu, \sigma^2)$
Student's t	ν (degree of freedom)	t_ν
Uniform	minimum and maximum	$\text{Uniform}(a, b)$
Binomial	n (trials) and p (probability of the event)	$\text{Binom}(n, p)$
Poisson	λ (mean rate)	$\text{Pois}(\lambda)$
Exponential	λ (rate)	$\text{Exponential}(\lambda)$
Gamma	α (shape), β (rate) or $\theta = 1/\beta$ (scale)	$\Gamma(\alpha, \beta)$
Beta	α, β (shape)	$B(\alpha, \beta)$

-parametric distributions are functions

Probability Mass Function (PMF)	-for a discrete random variable X , PMF is $p_X(x)$ such that $P(X=x) = p_X(x)$ -sum over all possible x is 1: $\sum_x p_X(x) = 1$
Probability Density Function (PDF)	-for a continuous random variable X -probability over an interval: $P(a \leq X \leq b) = \int_a^b f_X(x) dx$ -integral of PDF over entire line is 1: $\int_{-\infty}^{\infty} f_X(x) dx = 1$
Cumulative Distribution Function (CDF)	- $F_X(x) = P(X \leq x)$ -non-decreasing function: for $x_1 < x_2$, $F(x_1) \leq F(x_2)$

-properties of distributions

Normal	-bell-shaped, symmetrical around the mean
Uniform	-constant density over the interval
Binomial	-discrete distribution for # of successes in n Bernoulli trials -for large n , $\text{Binomial}(n, p)$ approximates $N(np, np(1-p))$
Poisson	-parameter: λ is both mean and variance -models count data over a fixed interval/time
Exponential	-one parameter of rate -often used for time between events
Gamma	-shape and scale (or rate = 1/scale) control shapes -models positive (skewed) data, can be flexible with different shapes
Beta	-location and scale parameters -supports values in $[0, 1]$, often for proportions/probabilities

3.5. Significance of Normal Distributions

Central Limit Theorem (CLT)	-sums/means of independently and identically distributed random variables $\rightarrow$ normal for large n -underpins Z-tests, t-tests, ANOVA, many parametric methods -many real-world phenomena approximate normality (e.g. measurement errors, biological variations)
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-note: even though the original distribution is uniform, the distribution of sample means is approximately normal when n is large

Week 4: t-test and Analysis of Variance

4.1. Test Procedure

-two ingredients: statistic and its theoretical distribution

-statistic represents what we want to know, meaning that it may differ contextually

- when we want to test whether two or more groups are different from one another, mean or variance could be used, but you might want to employ others

-once we know the theoretical distribution of the statistic, we can calculate the possible range of the statistic $\rightarrow$ confidence interval

Confidence Interval (CI)	-the range of the statistic that would lie in under repeated sampling -long run frequency of success $CI = \text{Estimate} \pm (\text{Standard Error}) \times (\text{Critical Value})$ -both the standard error and the critical value depend on the distributional assumptions -reading the lowest or highest end of the CI will provide whether the estimate of the statistic is statistically significant
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4.2. Identifying Differences

Chi-Square Distribution	-evaluates independence in categorical data or test variance -could be derived from Gamma distribution: $\chi^2(r) = Ga(\frac{r}{2}, \frac{r}{2})$ -representative case: a sample is drawn from a multinomial distribution / multiple categories $\chi^2 = \sum_i \frac{(O_i - Np_i)^2}{Np_i}$ where p_i is the theoretical probability of the category i -it arises as the distribution of a sum of squared standard normal variables -used for hypothesis tests involving: - independence tests (requires a contingency table) - goodness-of-fit tests (requires observed vs. expected frequencies) - variance testing (used with normally distributed data)
One-Way Table	=== One-Way Chi-Squared Goodness-of-Fit === Observed counts: [60 45 55 40] Expected counts: [50. 50. 50. 50.] Chi-Squared Statistic: 5.0000 -p-value is determined by $\chi^2_{df=3}$
Two-Way Case	-expected rate is calculated from the two margins: (row/column total) / (grand total) - expected proportion (row): 0.5, 0.5 - expected proportion (column): 0.2, 0.45, 0.55 -degree of freedom: $(p-1)(q-1)$ Table = [[5 20 35] [15 25 20]] Chi2 = 9.65, p = 0.008, dof = 2 Expected frequencies: [[10. 22.5 27.5] [10. 22.5 27.5]]

4.3. T-test

One Sample	-tests if mean of population differs from hypothesized μ_0 -statistic: $T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$
Two Sample (independent samples)	-compares means of two groups -(under equal variances) statistic: $T = \frac{\bar{X}_A - \bar{X}_B}{s_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}} \quad (s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1-1 + n_2-1)}})$ -if variances are unequal, use Welch's t-test
Student's t Distribution	-t distribution is derived from χ^2 distribution -considers the difference between the sample and population means in the scale of the sample standard deviation

	-t statistic is defined by the ratio between a variable following the standard normal distribution and one with χ^2 distribution, which are independent one another $W \sim \mathcal{N}(0, 1^2)$ $V \sim \chi^2(r)$ given $W \perp V$ $T = W/\sqrt{V/r}$ $T \sim t_{n-1}, \frac{(n-1)}{\sigma^2} s^2 \sim \chi_{n-1}^2$
Paired t-test	-used when same subjects measured twice -data reduce to differences $d_i = X_{i,post} - X_{i,pre}$ ☞ check if average difference $\bar{d}$ is zero: $T = \frac{\bar{d}}{s_d/\sqrt{n}}$ where $\bar{d} = \frac{1}{n} \sum d_i$

4.4. Analysis of Variance

-key question: do the different groups have statistically significantly different variances?

-the total variation is decomposed into between-group and within-group variation

One-Way ANOVA	$y_{ij} = \mu + \alpha_i + \epsilon_{ij}$ -μ is the grand mean, α_i is group level deviations from μ -note that α_i should be estimated ☞ could not be tested -$i = 1, \dots, k$ are group indexes, and j are observation indexes within group i
F-Statistic	-H_0: all group means are equal ($x_1 = \dots = x_G$) $F = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{\sum_{i=1}^K n_i (\bar{Y}_i - \bar{Y})^2 / (K-1)}{\sum_{i=1}^K \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2 / (N-K)}$ -significance is tested by the distribution of the statistics under H_0: $F \sim F_{K-1, N-K}$
Assumptions	-normality -homogeneity in variance -independence
Two-Way ANOVA	-used when two factors of interest are considered -the values are decomposed into the grand mean, the main effects (of the two factors), and the interaction between two factors $Y_{ijk} = \mu_{ij} + \epsilon_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk}$ -can be extended to multi-way ANOVA

4.5. Tukey's Ladder of Transformations

-data often deviate from normality or show heteroscedasticity

-Tukey's Ladder: set of power transformations to stabilize variance or approach normality

λ	-2	-1	-0.5	0	0.5	1	2
Transform	$1/x^2$	$1/x$	$1/\sqrt{x}$	x	$\sqrt{x}$	x	x^2
Skewness	Right	Right	Right	0	Left	Left	Left

-typical approach: inspect data distribution and skewness → try transformations from the ladder → choose the transform that yields more stable variance or improved normality

- skewness is related to the degree of three of deviations $\textcircled{\text{E}} \frac{\sum_{i=1}^N (Y_i - \hat{Y})^3 / N}{s^3}$
- kurtosis is related to the degree of four of deviations $\textcircled{\text{E}} \frac{\sum_{i=1}^N (Y_i - \hat{Y})^4 / N}{s^4} - 3$

Week 5: Linear Regression

5.1. Introduction

-terminology

Treatment (Predictor)	-a factor or variable we observe or manipulate to evaluate its effect
Effect	-the expected change in the outcome caused by changes in the predictor

→ goal: to estimate the causal effect of predictors on an outcome under certain assumptions

-ANOVA can be represented as a special case of regression with categorical predictors

→ linear regression extends ANOVA by modeling a continuous dependent variable as a function of one or more continuous/categorical independent variables (treatments)

-regression allows quantification of causal effects by adjusting for confounders

-interpretation of regression coefficients as effects requires assumptions like no omitted-variable bias and correct model specification

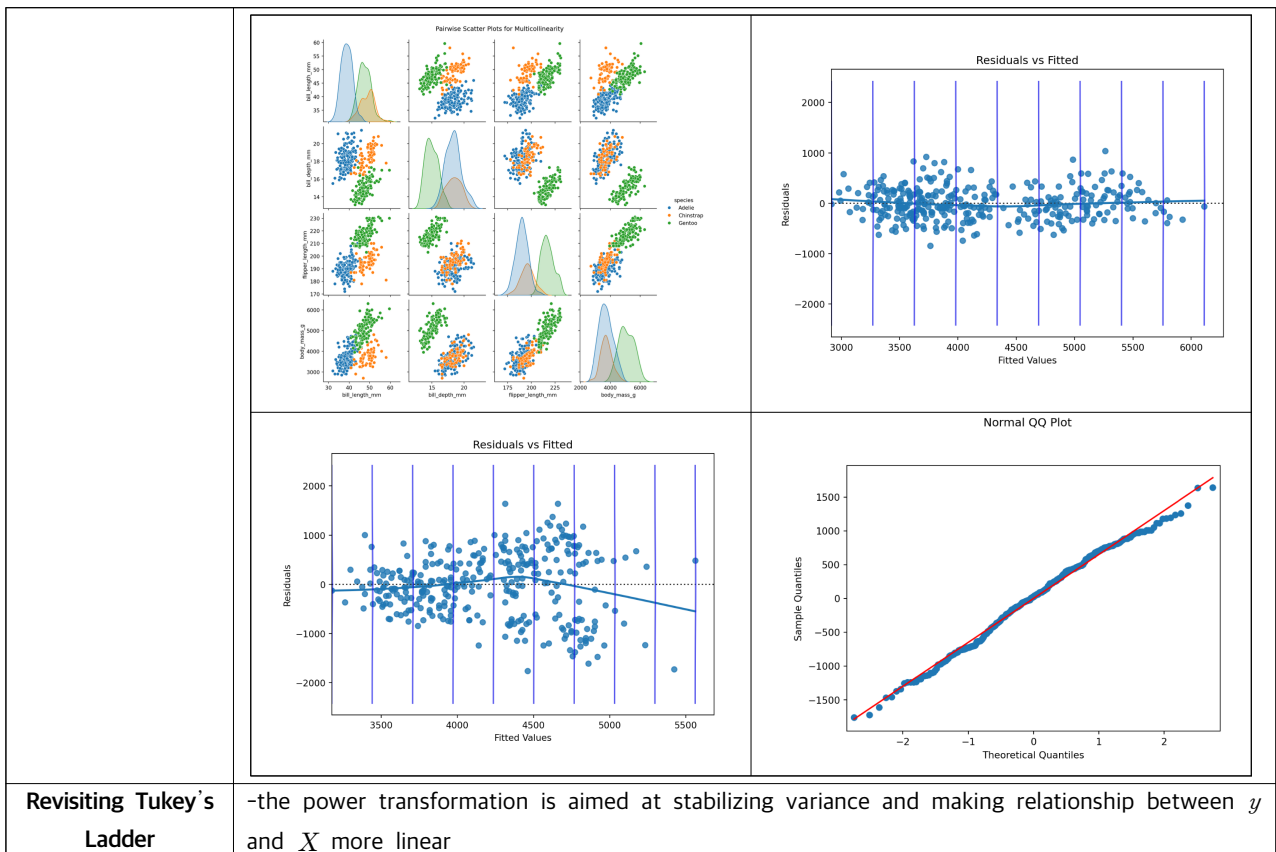
5.2. Linear Regression

Assumptions	Linearity	-the relationship between predictors and outcome is linear
	Independence	-observations are independent
	Homoscedasticity	-constant variance of residuals
	Normality	-residuals are normally distributed
	No Multicollinearity	-(for multiple regression) predictors should not be highly correlated
Simple Linear Regression Model	$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \epsilon_i \sim N(0, \sigma^2)$	
Multiple Linear Regression Model	$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \epsilon_i$ -estimate individual effects, controlling for other variables -useful for evaluating multiple treatments simultaneously	
Extensions	-dummy variables are used to represent categories -interactions between predictors can be included to model non-additive effects	
F-test	Ⓜ multiple regression model with 3 explanatory variables: $Y_i = \beta_0 + \sum_{i=1}^3 \beta_i X_i + \epsilon_i$ -F-test compares the residual sums of squares from the null model and a designed model -here we consider $H_0 = \beta_1 = \beta_2 = \beta_3 = 0$ • null model: $y_i = \beta_0 + \epsilon_i \rightarrow p_1 = 1$ • full model: $y_i = \beta_0 + \sum_{i=1}^3 \beta_i X_i + \epsilon_i \rightarrow p_2 = 4$ -F-statistic: $F = \frac{(\text{Explained variance}) / (p_2 - p_1)}{(\text{Residual variance}) / (n - p_2)}$ $= \frac{(\text{Total variance} - \text{Residual variance}) / (4 - 1)}{(\text{Residual variance}) / (n - 4)}$ $\sim F_{4-1, n-4}$	

Model Evaluation	-R-squared(R^2) is the ratio between the explained variance and the total variance of the outcome variable $R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$ -adjusted R^2 controls the number of explanatory variables: more explanatory variables penalizes the plain R^2 $R_{adj}^2 = 1 - \left(\frac{n-1}{n-1-k} (1 - R^2) \right)$ where k is the number of predictors, R^2 is the plain R-squared
Matrix Notation	$\mathbf{y} = \mathbf{X}\beta + \epsilon$ -$\mathbf{y}$: $n \times 1$ vector of responses -$\mathbf{X}$: $n \times (k+1)$ design matrix (the first column is all 1 for β_0) -β: $(p+1) \times 1$ vector of unknown coefficients -ϵ: $n \times 1$ vector of errors
Ordinary Least Squares (OLS)	-objective: minimize the sum of squared residuals $\arg \min_{\hat{\beta}} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \arg \min_{\hat{\beta}} \sum_{i=1}^n (y_i - \mathbf{X}_i \hat{\beta})^2 = \arg \min_{\hat{\beta}} (\mathbf{y} - \mathbf{X} \hat{\beta})^\top (\mathbf{y} - \mathbf{X} \hat{\beta})$ -$\hat{\beta}$ is calculated as $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ -geometrically, OLS is a projection of the outcome ($\mathbf{y}$) onto the predictor space (column space of $\mathbf{X}$)
Variance of Coefficients	-under standard assumptions ($\epsilon \sim N(0, \sigma^2 \mathbf{I})$), $\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ -regression coefficients are expected to be significantly different from zero $\Rightarrow$ t-test $t = \frac{\hat{\beta}_j}{\sqrt{\text{Var}(\hat{\beta}_j)}} \sim t_{n-p-1}$ -larger diagonal elements of $(\mathbf{X}^\top \mathbf{X})^{-1} \Rightarrow$ larger variances $\Rightarrow$ unstable estimates

5.3. Diagnostics

Independent Covariates	-when the predictors are completely independent, the effect of each predictor from the multiple regression model is the same as the effect from the single regression model -the coefficients from the multiple model can be interpreted as the effect of each predictor on the outcome, controlling for the other predictors -cumulative regression
Diagnostic Visualizations	-pairwise scatter plots: multicollinearity -residual plots: homoscedasticity (등분산성), linearity • expectation: points conditioned by a certain fitted value are distributed like a Gaussian -normal quantile-quantile plots: normality • expectation: residuals to be normally distributed $\Rightarrow$ dots in the QQ plot are expected to follow the diagonal line



Week 6: Dimension Reduction

6.1. Introduction

-high dimensional data often exhibit multicollinearity or strong correlations among predictors

-dimension reduction techniques help:

- summarize data structure
- visualize in fewer dimensions
- assist with regression by reducing collinearity

-spatial data and dimension reduction

- multicollinearity commonly occurs (e.g. correlated satellite bands)
- dimension reduction can help to improve interpretability

6.2. Principal Component Analysis (PCA)

-PCA finds new orthogonal axes(=principal components) that captures maximal variance in descending order

-suppose data matrix X is $n \times p$, and columns are mean-centered

-with eigen-decomposition, $\text{Cov}(\mathbf{X}) = \frac{1}{n-1} \mathbf{X}^T \mathbf{X} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$

- $\mathbf{\Lambda}$: $\text{diag}(\lambda_1, \dots, \lambda_p) \Rightarrow$ eigenvalues, sorted descending
- $\mathbf{V}$: orthogonal matrix of eigenvectors, $\mathbf{V}^T \mathbf{V} = \mathbf{I}$

-then the principal components are $\mathbf{Z} = \mathbf{XV}$ ($n \times p$)

- each column of $\mathbf{Z}$ is a principal component
- λ_i = variance explained by the i -th component

-notes

- PCA can be viewed as a linear transformation of the original data matrix X into a new space defined

by the eigenvectors of the covariance matrix

- the first principal component is the direction of the maximum variance, the second is orthogonal (independent) to the first and captures the next highest variance, and so on
 - the principal components are uncorrelated and ordered by the amount of variance they explain
 - the first few principal components often capture most of the variance in the data, allowing for dimensionality reduction by choosing m components instead of p
 - it is common to standardize the data (mean=0, variance=1) before applying PCA
- we can create biplot, where two components are chosen to display their scores in two-dimensional plot
- observations as points in PC space
 - loadings (contributions of original features) as arrows

Rotation Methods	-after PCA, we get orthonormal directions, but they may not be best for interpretability -varimax rotation tries to maximize the sum of variances of squared loadings (while orthogonality is maintained) $\Rightarrow$ makes loadings more sparse $\max_Q \sum_j \sum_i (Q_{ij}^2)^2 \text{ subject to } Q^T Q = I$ -procrustes or oblique rotation can also be used with less strict constraints: might fix certain loadings to zero or keep certain transformations minimal -rotation can yield principal axes more aligned with known environmental factors or measurement sets
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6.3. Multidimensional Scaling (MDS)

- attempts to preserve distances (or dissimilarities) among points in a low-dimensional space
- metric MDS uses Euclidean distances
- MDS can handle distance matrices directly $\Rightarrow$ especially useful for non-Euclidean or custom distances (☹)
- spatial problems, non-linear relationships)
- suppose we have distance matrix $D=[d_{ij}] \Rightarrow$ MDS finds points $\{z_i\}$ in m -dimensions that minimizes:

$$\min \sum_{i < j} [d_{ij} - \|z_i - z_j\|]^2$$

Z matrix	$-Z=[z_1, z_2, \dots, z_n]$: matrix of coordinates in the lower-dimensional surface (n objects, m dimensions) -solution to the optimization problem -the stress function measures how well the distances in the lower-dimensional space match the original distances $\text{Stress}(Z) = \sqrt{\frac{\sum_{i < j} [d_{ij} - \ z_i - z_j\]^2}{\sum_{i < j} d_{ij}^2}}$
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- PCA focuses on maximizing variance directions
- MDS tries to preserve pairwise distances

6.4. Dimension Reduction in Regression / Spatial Data Analysis

-dimension reduction solutions can help with regression problems: helps produce more stable estimates

PCA Regression	-use uncorrelated principal components as predictors -reduces dimensionality $\Rightarrow$ mitigates collinearity -possibly employ another rotation methods if interpretability is desired • 'which sets of variables are driving each principal component?': orthonormal rotation (☹) varimax)
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-spatial data and constraints

- in some advance contexts, we might impose certain diagonal or sparsity constraints on the rotation

matrix to keep interpretability (☞ Procrustes, oblique rotations)

- strict orthonormal constraints ensures components remain orthogonal
- diagonal constraints: we might restrict certain loadings or keep a block diagonal structure if we know certain variables belong to the same factor domain (commonly used in spatial factor analysis or regional grouping – 공간데이터 내 분류 방법과 관련된 constraint 부여)

Week 7: Introduction to Spatial Data

7.1. Spatial Data

-data with geographic location or position information (contains 'where' along with 'what')

-requires special data structures for efficient storage and analysis

-inherently multidimensional (at least two-dimensional: longitude, latitude – note that longitude is for X-axis and latitude is for Y-axis)

Aspatial Data	Spatial Data
-tabular data without spatial reference -data that can be represented in a table format	-includes explicit location component -geometries are stored in a coordinate reference system (CRS) -data that can be represented in a table format with geometry

7.2. Components of Spatial Data

- (1) geometry: the spatial component (points, lines, polygons)
- (2) attributes: descriptive properties associated with geometries
- (3) spatial reference system: coordinates system definition
- (4) topology: relationships between spatial features

-geometry

- also called spatial support
- the size, shape, and orientation of the data collection: OD(point)-1D(line)-2D(polygon)-3D(volume)
- used to determine the proper methods for analysis

Point Support	-represented by coordinate pairs $\mathbf{s}_i = (x_i, y_i)$ ☞ weather stations, hospitals, crime incidents, cafes, restaurants, etc
Line Support	-represented by a series of connected points in order $\mathbf{L}_i = \{(x_{i1}, y_{i1}), \dots, (x_{in}, y_{in})\}$ ☞ roads, rivers, gas pipelines, railroads, etc (note: all are centerlines)
Polygon Support	-enclosed areas of closed rings of coordinates $\mathbf{P}_i = \{(x_{i1}, y_{i1}), \dots, (x_{in}, y_{in}), (x_{i1}, y_{i1})\}$ ☞ administrative boundaries, cropland areas, etc

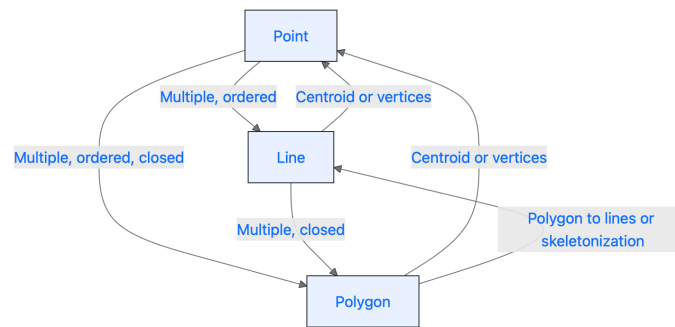
-spatial data representation with CRS

- coordinate reference system (CRS): a system that uses coordinates to establish the position of a point in space
- geographic coordinate system (GCS): a system that uses latitude and longitude to define locations on the earth's space
- projected coordinate system (PCS): a system that uses a flat, two-dimensional surface to represent the earth's curved surface
- spatial reference: the coordinate system and projection used to define the spatial location of features
- spatial extent: the geographic area covered by the data

7.3. Type Conversions

-geometries can be converted from one type to another type (appropriate type for analysis)

-called transformation or casting



- polygon to points: centroids
- polygon to lines: skeletonization, border extraction

7.4. Spatial Centrography (중심기술법)

-centrography

- center + graphy: study of the center
 - summarizes the distribution of points
- different meanings of centers from points

Mean Center	-multidimensional extension of the mean: average of x and y coordinates -may not be one of the data points -sensitive to outliers -the point that minimizes the sum of squared distances to all points $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$
Median Center	-the point that minimizes the sum of distances to all points -more robust to outliers than mean center -one of the data points
Standard Distance	-a measure of the dispersion of points around the mean center -spatial equivalent of standard deviation $SD = \sqrt{\frac{1}{n} \sum_{i=1}^n ((x_i - \bar{x})^2 + (y_i - \bar{y})^2)}$
Standard Deviation Ellipse (SDE)	-directional variant of standard distance -the axes of ellipse represent the standard deviations in the x and y directions and rotation angle (θ) -following the ellipse definition, the standard deviational ellipse contains major and minor axes as well as the angle of rotation

-nearest neighbor index (NNI)

- nearest neighbor distance: the distance from each point to its nearest neighbor
- useful for understanding the spatial distribution of points: measure of degree of clustering or dispersion of points

$$NNI = \frac{\bar{d}_{\text{observed}}}{\bar{d}_{\text{expected}}}$$

- where $\bar{d}_{\text{observed}} = \frac{1}{n} \sum_{i=1}^n \min_{j \neq i} d_{ij}$ and $\bar{d}_{\text{expected}} = \frac{1}{2\sqrt{n/A}}$ (n : number of points; A : area of the region)
- $\bar{d}_{\text{expected}}$: average nearest neighbor distance of a random distribution of points, i.e. Poisson process)
- $NNI < 1$: clustering; $NNI = 1$: random distribution; $NNI > 1$: dispersion

7.5. Visual Exploration of Spatial Data

-rule-of-thumb approach to understanding spatial data

-two approaches

Choropleth Maps and Dot Maps with Hue	-choropleth map: a thematic map where areas are shaded or patterned in proportion to the value of a variable being represented -dot map: a map where points are used to represent the presence of a feature or phenomenon -hue: a color attribute that represents the value of a variable -useful when one variable is concerned
Division of Space into Regions by Dimensions	-a method to visualize the relationship between two variables by dividing the space into regions based on coordinate ranges -steps • prepare the geometries (centroids if polygons are used) • divide the space into regions based on quantiles or equal intervals • create a scatterplot for each region while preserving the relative position of the regions

Week 8: Distance and Neighbor Search

8.1. Why Spatial Matters

-Tobler's First Law of Geography: "everything is related to everything else, but near things are more related than distant things"

-consideration of location: location creates spatial dependence and spatial heterogeneity

-traditional statistics may:

- miss spatial patterns
- violate independence assumptions
- lead to incorrect inferences

8.2. Distance Calculations

Euclidean Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ -straight line distance -assume a flat surface: commonly used in projected coordinate systems, does not work well with latitude/longitude -commonly used for small study areas
Manhattan Distance	$d = x_2 - x_1 + y_2 - y_1 $ -sum of absolute differences -useful in grid-based or urban settings -also called taxicab or city block distance
Great Circle Distance	-for points with (longitude, latitude) (ϕ_1, λ_1) and (ϕ_2, λ_2): $d = r \times \arccos(\sin\phi_1\sin\phi_2 + \cos\phi_1\cos\phi_2\cos(\lambda_2 - \lambda_1))$ -accounts for earth's curvature(r) -important for larger study areas -r = earth's radius ($\simeq 6378.137\text{km}$)
Network Distance	-distance based on travel through a network (🗺️ roads) -measures the shortest path along edges • path: sequence of connected nodes • node: intersection or endpoint • edge: connection between two nodes -can incorporate: direction constraints (one-way), impedance (travel time, traffic), capacity limitations -often significantly different from straight-line distance (detour index: network/Euclidean > 1)

	-notes on the distances in urban road networks • shape of the network affects the ratio • network distance is often much longer than Euclidean distance: perpendicular road networks can lead to the approximate ratio up to $\sqrt{2}:1$ • note that network distance often assumes that origin and destination are snapped to the nearest road segment
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-distance between complex geometries

- point to line: shortest perpendicular distance
- point to polygon: distance to nearest edge/vertex
- line to line: minimum distance between any two points
- polygon to polygon: minimum distance between boundaries

8.3. Spatial Neighborhoods

-contiguity based neighborhoods

- polygonal geometries

Rook Contiguity	-shares edge
Queen Contiguity	-shares edge or vertex

-distance based neighborhoods

k-Nearest Neighbors	- k closest features -algorithm • calculate distances to all other locations • rank from nearest to farthest • select k nearest locations $\Rightarrow$ define as neighbors -advantage: ensures each location has exactly k locations $\Rightarrow$ works well with irregular distributions
Distance Band	-all features within specified distance

8.4. Spatial Weight Matrices

-spatial weight matrix represent spatial relationships, derived from the spatial data: $D(s) \rightarrow \mathbf{W}$

-each element w_{ij} quantifies the “(weighted) connection” between locations i and j

-usually row-standardized so each row sums to 1: $w_{ij}^* = \frac{w_{ij}}{\sum_j w_{ij}}$

- ensures comparability across observations
- facilitates interpretation as weighted average

-foundation for spatial autocorrelation measures and spatial models

-types of spatial weight matrices

Binary Weight Matrix	-for contiguity or threshold distance $w_{ij} = \begin{cases} 1 & \text{if locations } i \text{ and } j \text{ are neighbors} \\ 0 & \text{otherwise} \end{cases}$ -usually $w_{ii} = 0$ (no self-neighbors) -simple but loses information about relative proximity
Inverse Distance Weight Matrix	-based on distance decay: $w_{ij} = \begin{cases} \frac{1}{d_{ij}^p} & \text{if } d_{ij} < \text{threshold} \\ 0 & \text{otherwise} \end{cases}$ - p is the power parameter (often 1 or 2)

	-closer features have higher weights -more realistic than binary weights
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-symmetry of spatial weight matrices

- binary contiguity: usually symmetric
- k-nearest neighbor: usually asymmetric
- distance-based: symmetric with fixed distance bands (may be asymmetric with adaptive bandwidths)

-handling asymmetry

- symmetry matters for certain statistical properties and eigenvalue decomposition
- row-standardization typically creates asymmetry
- force symmetry: 1) $w_{ij} = 1$ if either i is a neighbor of j or j is a neighbor of i ; 2) $\max(w_{ij}, w_{ji})$

8.5. Applications of Spatial Weight Matrices

Weight Matrix from Points	-calculate all pairwise distances -apply distance threshold or k-nearest criterion -compute weights (binary, inverse distance, etc) -spatial clustering affect weights
Weight Matrix from Lines	-used as a background for point data -consider shortest distances between lines leveraging connectivity or network relationships -relevant for roads, rivers, transportation networks -may incorporate directionality (flow)
Weight Matrix from Polygons	-contiguity (Rook/Queen) -centroid-based distance: various centroid calculations $\textcircled{A}$ inside the polygon, average of all vertices -most common in administrative units -shared boundary length weighting possible • shared boundary length: may proxy for the strength of interaction between two polygons $w_{ij} = \frac{L_{ij}}{\sum_{k=1}^n L_{ik}}$ • L_{ij}: length of the shared boundary between polygons i and j • L_{ik}: length of the shared boundary between polygon i and all other polygons k • n: number of polygons

Week 9: Spatial Sampling and Point Pattern Analysis

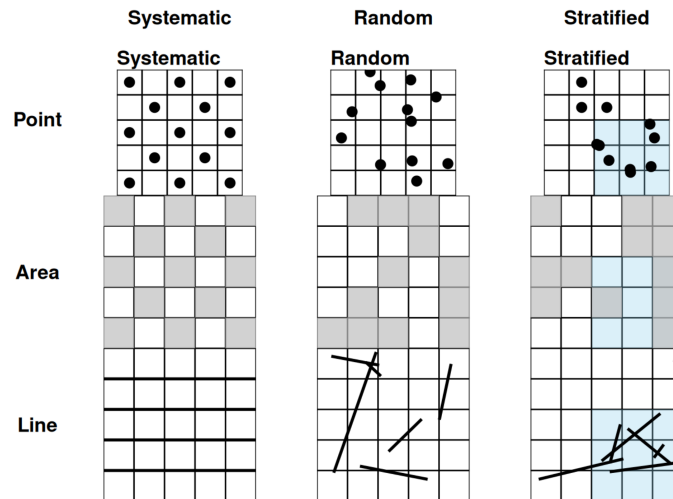
9.1. Spatial Sampling

-spatial sampling designs

- define an observed point pattern as a matrix of coordinates

$$S = \begin{bmatrix} s_{11} & s_{12} \\ \vdots & \vdots \\ s_{n1} & s_{n2} \end{bmatrix}$$

	Matrix View	Good for
Simple Random	- $S \sim U(W)$ where W : 분석 대상 공간	-baseline surveys
Systematic Grid	-lattice rows/cols	-equal coverage
Stratified	-block-diagonal frame	-heterogeneous regions
Adaptive Cluster	-separate S_i from S_0 • S_i: intensive sampling area • S_0: non-intensive sampling area	-soil contaminant sampling in suspected areas



9.2. Complete Spatial Randomness (CSR)

- CSR is a null hypothesis for point patterns $\Rightarrow$ Poisson process with intensity λ
- no interaction between points & no clustering or regularity
- independently distributed
- equally likely to occur anywhere in the region
- have constant intensity across space

Poisson Point Process	$P\{N(A) = k\} = \frac{(\lambda A)^k e^{-\lambda A }}{k!}$ -$N(A)$: the number of points in area A -λ: the intensity (average number of points per unit area) -A: the area of A
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9.3. Quadrat Test

- compares the observed point pattern to CSR
- divides study area into equal-sized cells (quadrats) and counts points in each quadrat
- under CSR, count follow Poisson distribution
- chi-square test assesses deviation from expected counts

Quadrat Test	-partition W into m cells -let count vector $\mathbf{n}$ -$\hat{\lambda} = \frac{n}{ W }$, $\mu = \hat{\lambda}\mathbf{a}$ ($\mathbf{a}$는 단위구역의 면적) $\Rightarrow$ test statistic: $\chi^2 = (\mathbf{n} - \mu)' \text{diag}(\mu)^{-1} (\mathbf{n} - \mu)$
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-limitations

- quadrat size and configurations may affect the result
- unable to capture the interaction (quadrats are independent)
- less powerful than distance-based methods (less continuous)

9.4. Summary Functions for Point Patterns

-five major functions

- F: empty space function
- G: nearest neighbor distance function
- J: combines F and G functions

- K: Ripley's K-function
- L: normalized version of K-function

	Definition	Formula	Clustered Pattern if
F-function	-measures distance from random points to their nearest observed point	- $F(r) = P(\min_i (d_{\text{random points}, i}) \leq r)$ - $F(r) = 1 - \exp(-\lambda\pi r^2)$ under CSR	-empirical < theoretical
G-function	-measures distance from each point to its nearest neighbor point • $G(\max r) = 1$	- $G(r) = P(\text{distance to nearest neighbor} \leq r)$ - $G(r) = 1 - \exp(-\lambda\pi r^2)$ under CSR	- empirical > theoretical
J-function	-combines F and G functions • less affected by edge effects than F, G	- $J(r) = \frac{1 - G(r)}{1 - F(r)}$ - $J(r) = 1, \forall r$ under CSR	-empirical < theoretical ($J(r) < 1$)
K-function	-expected number of points within distance r of an arbitrary point • sensitive to patterns at multiple scales	- $K(r) = \frac{1}{\lambda} E[\text{number of points within } r]$ - $K(r) = \pi r^2$ under CSR	- empirical > theoretical
L-function	-normalized version of K	- $L(r) = \sqrt{\frac{K(r)}{\pi}}$ - $L(r) = r$ under CSR	- empirical > theoretical ($L(r) - r > 0$)

-Monte-Carlo envelope algorithm: utilized to evaluate the significance of observed functions

- simulate B CSR patterns
- compute $K_b(r)$ for each, where b stands for b -th simulation set
- upper/lower envelopes = max, min across b

9.5. Kernel Density Estimation (KDE)

-non-parametric way to estimate the probability function of a random variable

-distances from grid G (size N) to points: $\mathbf{D}_G$ ($N \times n$ matrix) $\Rightarrow [\mathbf{D}_G]_{ki} = d(s_k, s_i)$

-estimator:

$$\hat{\lambda} = \frac{1}{h^2} \phi\left(\frac{\mathbf{D}_G}{h}\right) \mathbf{1}_n$$

- ϕ : kernel function (models distance decay)
- h : bandwidth $\Rightarrow$ higher h will yield smoother estimates
- $\mathbf{1}_n$: a vector of ones of length n

Week 11: Spatial Autocorrelation

11.1. Autocorrelation

-correlation: a measure of the linear relationship between two variables

$$\rho = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{\text{Cov}(X, Y)}{\rho_X \rho_Y}$$

-autocorrelation: correlation of a variable with itself over space or time

-spatial autocorrelation could be different by the definition of a neighborhood (i.e. spatial weights matrix)

- spatial weights matrix: binary / real values, row-standardized

Data Types	Distance	Neighbors
point	euclidean	KNN, distance bands
line	euclidean	topology
polygon	contiguity	rook, queen, KNN (centroids)
raster	euclidean	rook, queen, KNN (centroids)

11.2. Global Spatial Autocorrelation Measures

Moran's I	$I = \frac{n}{\sum_i \sum_j w_{ij}} \frac{\mathbf{z}^T \mathbf{W} \mathbf{z}}{\mathbf{z}^T \mathbf{z}} = \frac{n}{S_0} \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$ - $S_0 = \sum_i \sum_j w_{ij}$ $\Rightarrow$ controls the density of the weights matrix - range: generally between -1 and 1 $I > 0$: positive spatial autocorrelation (clustering) $I = 0$: no spatial autocorrelation (random) $I < 0$: negative spatial autocorrelation (dispersion) - statistical testing - null hypothesis: no spatial autocorrelation - expected value under null: $E[I] = -\frac{1}{n-1}$ - inference through permutation, normality assumption, or Monte Carlo simulation
Geary's C	$C = \frac{n-1}{2S_0} \frac{\sum_i \sum_j w_{ij} (x_i - x_j)^2}{\sum_i (x_i - \bar{x})^2}$ - focusing on squared differences between pairs - range: 0 to approximately 2 $C < 1$: positive spatial autocorrelation (clustering) $C = 1$: no spatial autocorrelation (random) $C > 1$: negative spatial autocorrelation (dispersion)
Getis-Ord's G	$G = \frac{\sum_i \sum_j w_{ij} x_i x_j}{\sum_i \sum_j x_i x_j}$ - uses raw values - focuses on concentration of high or low values - expected value under random pattern: $E[G] = \frac{\sum_i \sum_j w_{ij}}{n(n-1)}$ - interpretation $G > E[G]$: clustering of high values $G = E[G]$: no clustering pattern $G < E[G]$: clustering of low values
Permutation Test	- procedure - shuffle data values (without replacement) with the same spatial configuration of the data - calculate the test statistic for each permutation - compare the observed statistic with the distribution of permuted statistics - pseudo p-value: proportion of permuted statistics greater than or equal to the observed statistic

11.3. Local Spatial Autocorrelation Measures

- global to local: decompose global statistics into contributions from individual locations

-allows identification of:

- clusters (hot spots, cold spots)
- spatial outliers
- areas contributing most to global autocorrelation

Local Moran's I_i	$I_i = \frac{(x_i - \bar{x}) \sum_j w_{ij} (x_j - \bar{x})}{\sum_j (x_j - \bar{x})^2 / n}$ -sum of all local Moran's I_i is proportional to global Moran's I $\sum_i I_i = \frac{n}{S_0} \times \text{global Moran's } I$ -Moran scatterplot • X-axis: values of variable at location i • Y-axis: spatially lagged value of variable at location i -interpreting local Moran's I_i • positive I_i: location i surrounded by similar values (cluster) <table border="1" data-bbox="440 835 1407 898"> <tr> <td>High-High (HH)</td><td>-high value surrounded by high values → 1st quadrant</td></tr> <tr> <td>Low-Low (LL)</td><td>-low value surrounded by low values → 3rd quadrant</td></tr> </table> • negative I_i: location i surrounded by dissimilar values (outlier) <table border="1" data-bbox="440 952 1407 1014"> <tr> <td>High-Low (HL)</td><td>-high value surrounded by low values → 4th quadrant</td></tr> <tr> <td>Low-High (LH)</td><td>-low value surrounded by high values → 2nd quadrant</td></tr> </table>	High-High (HH)	-high value surrounded by high values → 1 st quadrant	Low-Low (LL)	-low value surrounded by low values → 3 rd quadrant	High-Low (HL)	-high value surrounded by low values → 4 th quadrant	Low-High (LH)	-low value surrounded by high values → 2 nd quadrant
High-High (HH)	-high value surrounded by high values → 1 st quadrant								
Low-Low (LL)	-low value surrounded by low values → 3 rd quadrant								
High-Low (HL)	-high value surrounded by low values → 4 th quadrant								
Low-High (LH)	-low value surrounded by high values → 2 nd quadrant								
Local Getis-Ord's G_i^*	$G_i^* = \frac{\sum_j w_{ij} x_j}{\sum_j x_j}$ -identifies spatial clustering of high (hot-spots) or low values (cold-spots) -includes value at location i in the calculation								

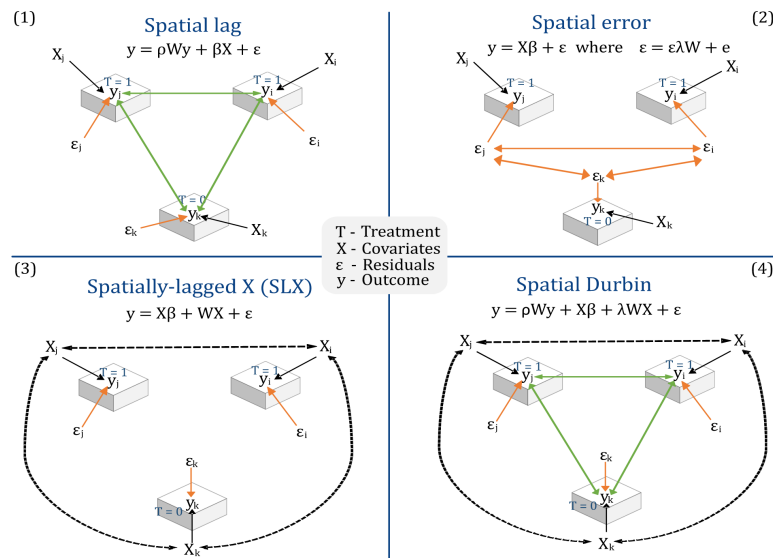
11.4. Recent Advances in Moran's I

Adjusted Moran's I for Heteroscedasticity	-modified version $I_{\text{adj}} = \frac{n}{S_0} \frac{\sum_i \sum_j w_{ij} \frac{(x_i - \bar{x})(x_j - \bar{x})}{\sigma_i \sigma_j}}{\sum_i \frac{(x_i - \bar{x})^2}{\sigma_i^2}}$ • σ_i: standard deviation at location i
Bivariate Moran's I	-measures whether the value of a variable x at location i is related to the spatial lag of a different variable y around that location $I_{xy} = \frac{n}{S_0} \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(y_j - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_j (y_j - \bar{y})^2}}$ $I_{xy,i} = z_{x,i} \sum_j w_{ij} z_{y,j} \text{ where } z_{x,i} = \frac{x_i - \bar{x}}{s_x}, z_{y,j} = \frac{y_j - \bar{y}}{s_y}$ -interpretation: for a positive significant $I_{xy,i}$, • High-High: high x_i, high lagged y • Low-Low: low x_i, low lagged y

Week 12. Spatial Regression

12.1. Overview of Spatial Regression Models

-used when some of the OLS assumptions are violated due to spatial dependence



	Violation	Relevance	Formulation
Spatial Lag (SLM)	-exogeneity	-y	$y = \rho W y + X\beta + \epsilon$
Spatial Error (SEM)	-homoscedasticity	- ϵ	$y = X\beta + u$ where $u = \lambda W u + \epsilon$
Spatially Lagged X (SLX)		-X	$y = X\beta + WX\theta + \epsilon$
Spatial Durbin (SDM)	-exogeneity & homoscedasticity	-y and X	$y = \rho W y + X\beta + WX\theta + \epsilon$

-model evaluation: goodness of fit

pseudo-R^2	$\text{pseudo-}R^2 = 1 - \frac{\log - \text{likelihood of the model}}{\log - \text{likelihood of the null model}}$ -R^2 for spatial models -higher is better
log-likelihood	-the probability of observing data given the model parameters -higher is better
AIC	$AIC = -2 \cdot \log - \text{likelihood} + 2k$ -Akaike information criterion -lower is better
BIC	$BIC = -2 \cdot \log - \text{likelihood} + k \cdot \log(n)$ -Bayesian information criterion -lower is better

-interpreting the results

Spatial Lag (SLM)	$y = \rho W y + X\beta + \epsilon$ -ρ: spatial autoregressive parameter $\rightarrow$ strength of spatial dependence of the outcomes -all parameters estimated simultaneously
Spatial Error (SEM)	$y = X\beta + u$ where $u = \lambda W u + \epsilon$ -the error term u is modelled as spatially dependent -accounts for spatial dependence in errors -useful when the OLS residuals are spatially autocorrelated
Spatially Lagged X (SLX)	$y = X\beta + WX\theta + \epsilon$ -$WX\theta$: spatially lagged independent variable -θ: parameter for the spatially lagged independent variable

Spatial Durbin (SDM)	$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ -combines spatial lag and error models -allows for spatially lagged independent variables -useful for understanding spillover effects -can be used both for cross-sectional and time-series data
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12.2. Notes

-in SLM and SDM, direct and indirect effects can be estimated

- direct effect: the effect of a unit change in the independent variable on the dependent variable at the same location
- indirect effect: the effect of a unit change in the independent variable on the dependent variable at neighboring locations (= spillover effect)

-users should carefully choose the model based on the research question and data

- parsimonious model is preferred → SLM is preferred over SDM
- however it does not necessarily mean that the SLM is the best model for all cases

Week 13. Geographically Weighted Regression

13.1. Spatial Non-Stationarity

-traditional regression assumes spatial stationarity

- no spatial heterogeneity in relationships
- may miss local patterns and processes

-factors contributing to spatial non-stationarity

- environmental gradients
- local economic conditions / market segmentation
- cultural / policy differences

→ Tobler's First Law revisited: spatial scale matters for parameter estimation!

13.2. Geographically Weighted Regression

-core idea: allow regression parameters to vary smoothly across space

- GWR is a local regression (basically fitting n local models) → allows for spatially varying relationships between dependent and independent variables
- estimates parameters for each observation, capturing local variations
- useful when the relationships between variables is not constant across space
- can reveal how the influence of independent variables on the dependent variable change geographically

-GWR model can be expressed as:

$$y_i = \beta_0(u_i, v_i) + \sum_{j=1}^p \beta_j(u_i, v_i) x_{ij} + \epsilon_i$$

- y_i : dependent variable for observation i
- $\beta_0(u_i, v_i)$: intercept that varies by location (u_i, v_i)
- $\beta_j(u_i, v_i)$: coefficients for independent variables x_{ij} that vary by location
- ϵ_i : error term for observation i

-in matricial form,

$$\hat{\beta}(u_i, v_i) = (\mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{y}$$

- $\mathbf{W}(u_i, v_i)$: spatial weighting matrix → location-specific

- $\mathbf{X}$: design matrix (including the intercept)
- $\mathbf{y}$: response vector

-mathematical properties

Hat Matrix (S)	$\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$ $\mathbf{S} = \mathbf{X}(\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}$ where $\mathbf{W}$ is the stack of spatial weights for each observation, $\mathbf{W}_i$ -projection matrix that maps observed values to fitted values -measure of influence of each observation on the fitted values
Effective Number of Parameters	$n_{\text{eff}} = \text{trace}(\mathbf{S})$ -proportional to the sum of distance decay weights
Degree of Freedom	$\text{df}_{\text{eff}} = n - \text{trace}(\mathbf{S})$ where n is the number of observations

13.3. Spatial Weighting Functions

$-\mathbf{W}(u_i, v_i)$ is determined by spatial weighting function that decays with distance from the location (u_i, v_i)

Gaussian Kernel	$w_{ij} = \exp\left(-\frac{d_{ij}^2}{2h^2}\right)$
Bi-Square Kernel	$w_{ij} = \begin{cases} \left(1 - \left(\frac{d_{ij}}{h}\right)^2\right)^2 & \text{if } d_{ij} < h \\ 0 & \text{otherwise} \end{cases}$
Epanechnikov Kernel	$w_{ij} = \begin{cases} \frac{3}{4} \left(1 - \left(\frac{d_{ij}}{h}\right)^2\right) & \text{if } d_{ij} < h \\ 0 & \text{otherwise} \end{cases}$

- d_{ij} : distance between locations i and j
- h : bandwidth (distance threshold)
- w_{ij} : weighted assigned to observation j in the local regression at location i

-the choice of kernel affects the smoothness and locality of the regression estimates

-summary of kernel functions

	Support	Smoothness	Characteristics
Gaussian	infinite	smooth	commonly used
Exponential	infinite	smooth	moderate
Bi-Square	finite	smooth	smooth at boundary, local influence
Tri-Cube	finite	very smooth	smoother than bi-square, local influence

- infinite support kernels allow influence from all points whereas finite support kernels drop to zero beyond the bandwidth
- bandwidth controls the strength of the influence of distant points
- finite support kernels are more localized and may gain fitting speed with many zeros in the weight matrix

13.4. Bandwidth Selection

Fixed Bandwidth	-constant distance h -good for regular point patterns -may include varying number of neighbors
Adaptive Bandwidth	-fixed number of nearest neighbors -adjusts to point density

	-effective when spatial units are sparsely distributed - $h_i = d_{i,k}$ where k is the number of neighbors
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-selection methods

Cross-Validation (CV)	$CV = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$ - $\hat{y}_i$: predicted value for observation i without using its own value
AICc	$AICc = n \ln(\sigma^2) + n \ln(2\pi) + n \left\{ \frac{n + \text{trace}(\mathbf{S})}{n - 2 - \text{trace}(\mathbf{S})} \right\}$ -penalized model complexity (i.e. small bandwidth)
BIC	$BIC = n \ln(\hat{\sigma}^2) + n \ln(2\pi) + \ln(n) \text{trace}(\mathbf{S})$

13.5. Post-Estimation Diagnostics

-validity of spatial heterogeneity assumptions: check if parameters vary significantly across space

- selected bandwidth $h \simeq n$ implies that the relationship is close to global

-goodness-of-fit: AICc, BIC, pseudo R^2 , RSS

-residuals: check for spatial autocorrelation in residuals $\Rightarrow$ standardized residuals should be randomly distributed with no extreme values

13.6. GWR vs. Spatial Econometrics

	GWR	Spatial Econometrics
Parameter Variation	local (spatially varying)	global (constant)
Spatial Dependence	implicit through weights	explicit through $\mathbf{W}\mathbf{y}$ or $\mathbf{W}\mathbf{X}$
Focus	local relationships	global spatial processes
Estimation	weighted least squares	maximum likelihood
Interpretation	local effects	global effects + spillovers
Preferred When..	-interested in local parameter variation -relationship heterogeneity suspected -exploratory spatial analysis -local policy implications	-global spatial processes of interest -spillover effects important -causal inference focus -strong theoretical spatial model

13.7. Multiscale Geographically Weighted Regression (MGWR)

-problem with GWR

- single bandwidth for all variables $\Rightarrow$ assumes all relationships vary at same spatial scale
- may over-smooth or under-smooth different processes

-MGWR allows for

- variable-specific bandwidths
- different spatial scales for different processes $\Rightarrow$ more flexible modeling framework

-mathematical formulation

$$y_i = \sum_{k=0}^p \beta_k(u_i, v_i, h_k) x_{ik} + \epsilon_i$$

- h_k : bandwidth for variable k

-local estimation

$$\hat{\beta}_k(u_i, v_i, h_k) = (\mathbf{X}^\top \mathbf{W}_k(u_i, v_i, h_k) \mathbf{X}_k)^{-1} \mathbf{X}_k^\top \mathbf{W}_k(u_i, v_i, h_k) \mathbf{y}$$

-bandwidth selection: back-fitting algorithm

1. initialize all bandwidths
2. for each variable k :

- fix other bandwidths
- optimize h_k using partial residuals

3. repeat until convergence

☞ AICc is the usual criterion for multiple bandwidth selection

13.8. Model Selection Guidelines

-consider using GWR when:

- exploratory analysis: understanding spatial heterogeneity
- local relationships: interest in place-specific effects
- non-stationarity evidence: statistical tests suggest spatial variation

-think twice before applying GWR when:

- small samples: $n < 100$
- global processes: theory suggests uniform relationships
- high dimensionality: many predictors relative to sample size (may lead to overfitting)
- causal inference: spillover effects are of primary interest

13.9. Extensions of GWR

-temporal extension: geographically and temporally weighted regression (GTWR)

$$y_i = \sum_{k=0}^p \beta_k(u_i, v_i, t_i) x_{ik} + \epsilon_{it}$$

-machine learning extensions: geographically weighted models

- GWR with regularization: GW Ridge Regression, GW Lasso Regression, GW Elastic Net
- GW neural network: (deep) learning for complex spatial relationships, convolutional layers for spatial structure, attention mechanisms for adaptive weighting

13.10. Issues & Limitations

-scalability issues: computational complexity ☞ $O(n^3)$ for each location

- solutions: fast GWR algorithms, sparse matrix techniques, parallel computing, approximation methods

-bandwidth selection sensitivity

-multiple testing issues (false positives due to local parameters)